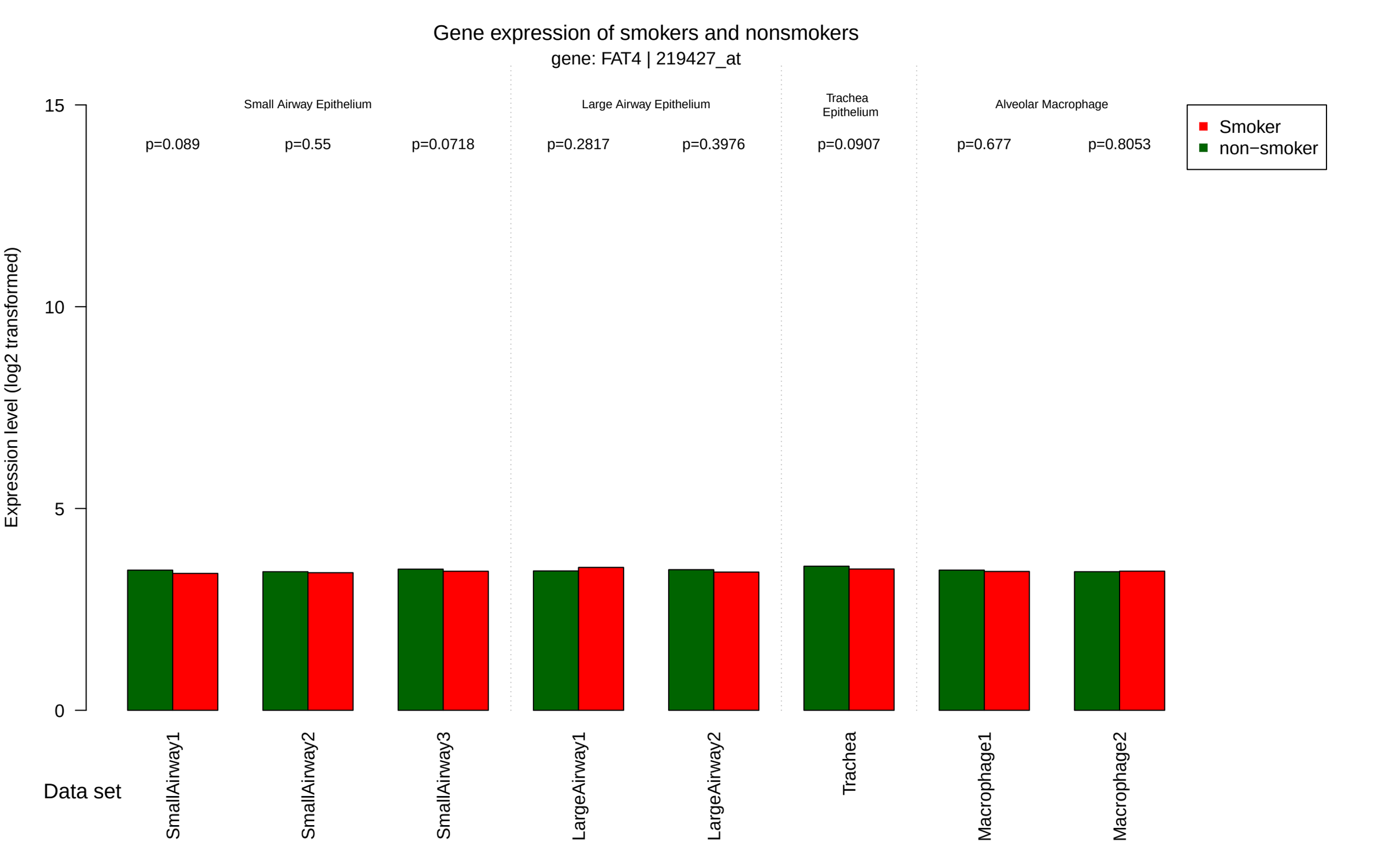
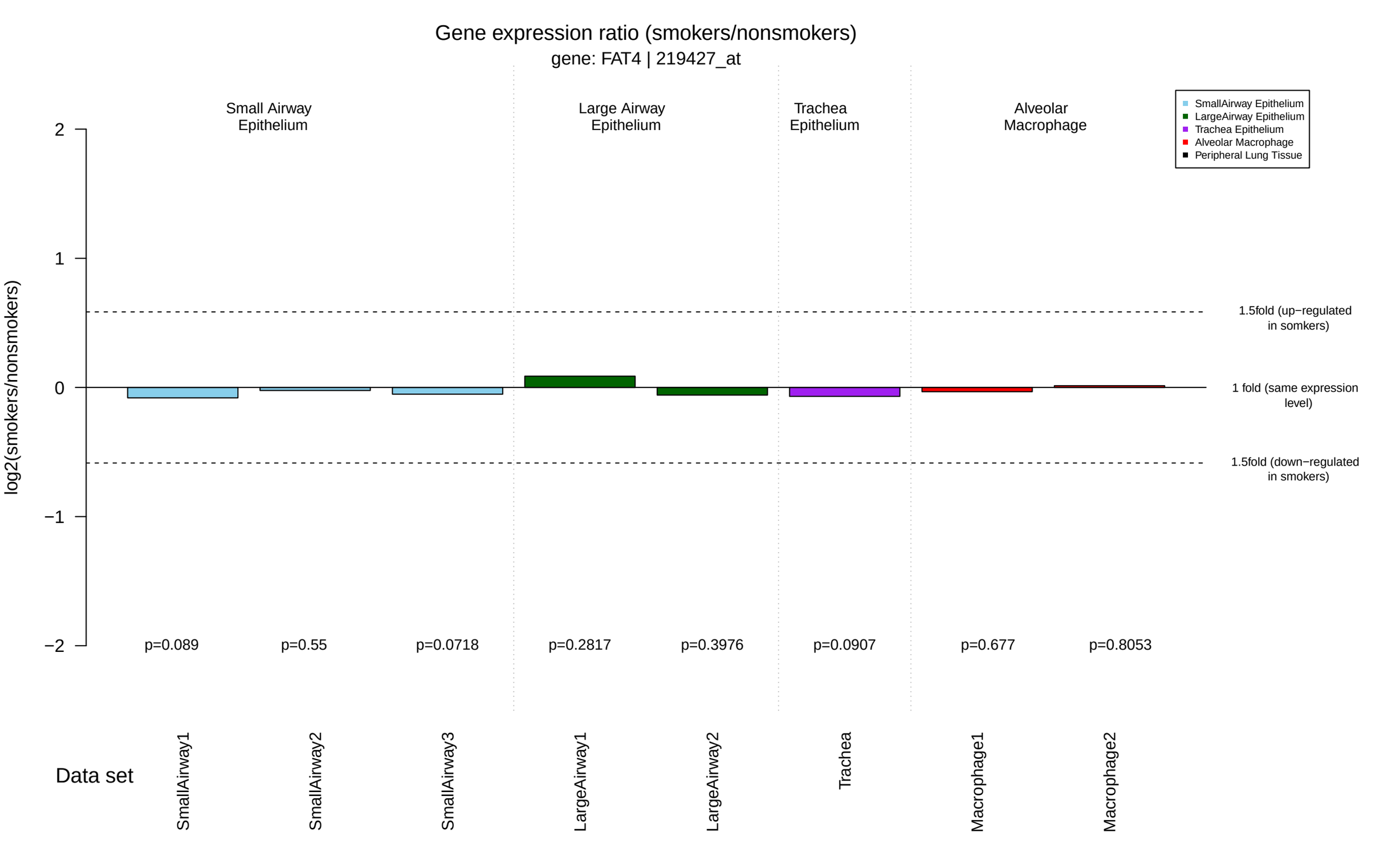
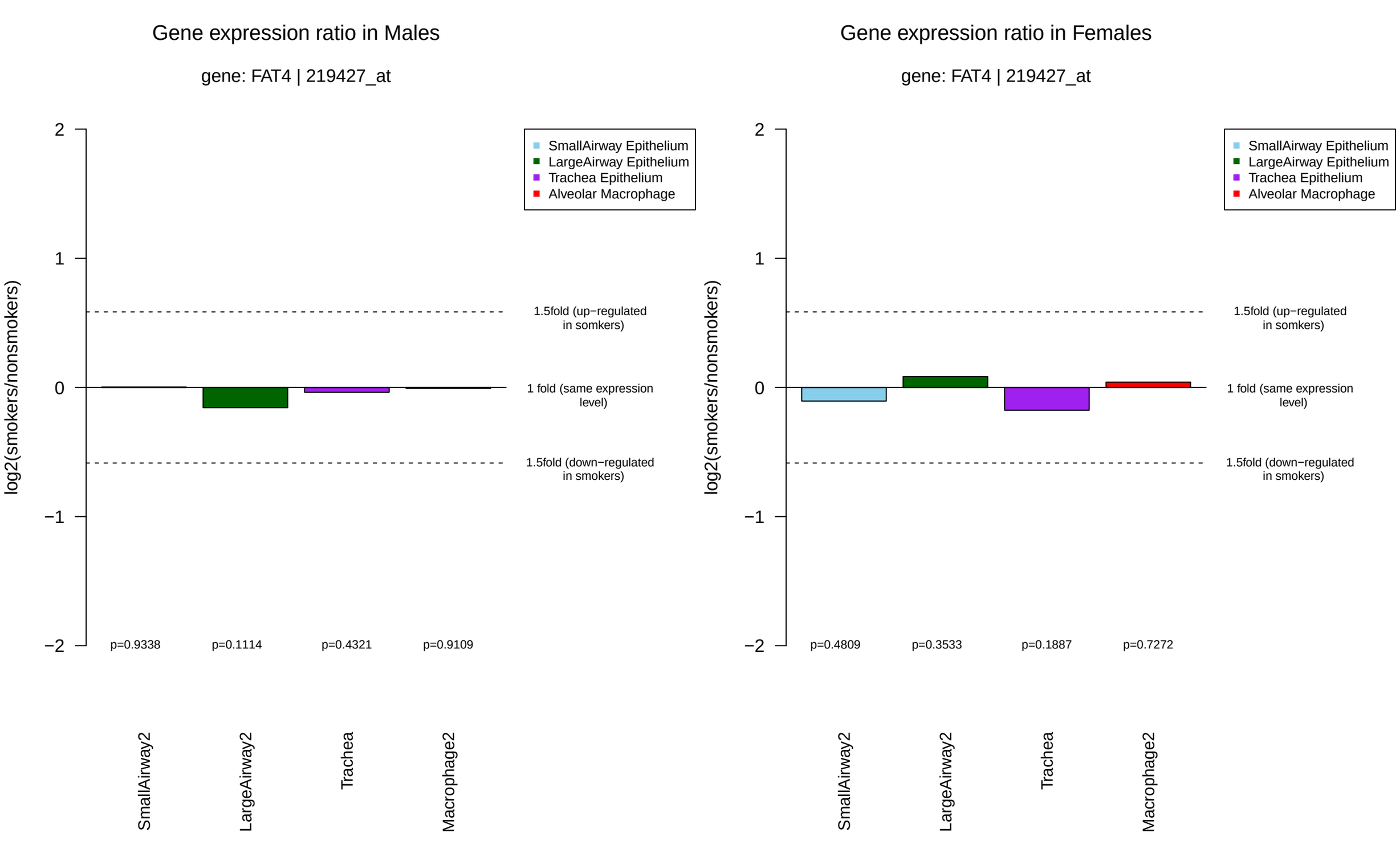
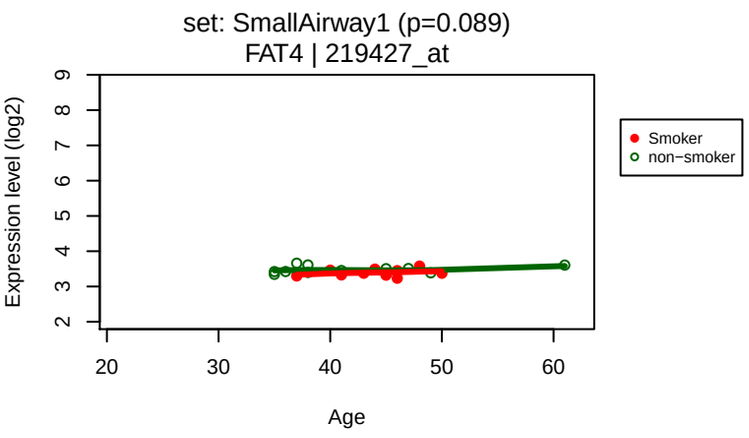
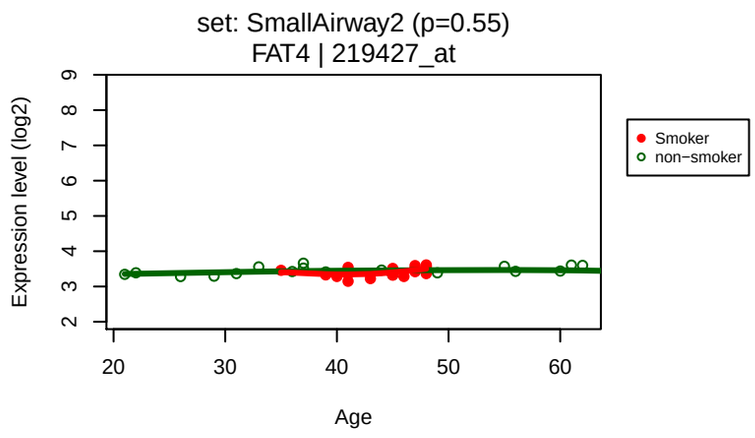
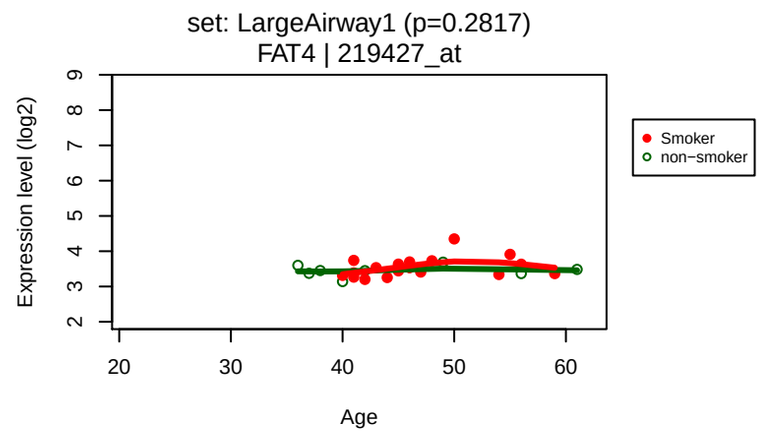

Gene expression of smokers and nonsmokers
gene: FAT4 | 219427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.089
p=0.55
p=0.0718
p=0.2817
p=0.3976
p=0.0907
p=0.677
p=0.8053
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

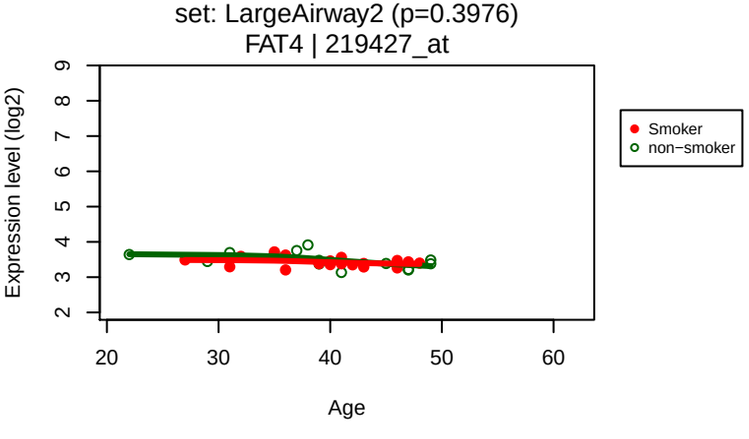

Gene expression ratio (smokers/nonsmokers)
gene: FAT4 | 219427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.089
p=0.55
p=0.0718
p=0.2817
p=0.3976
p=0.0907
p=0.677
p=0.8053
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

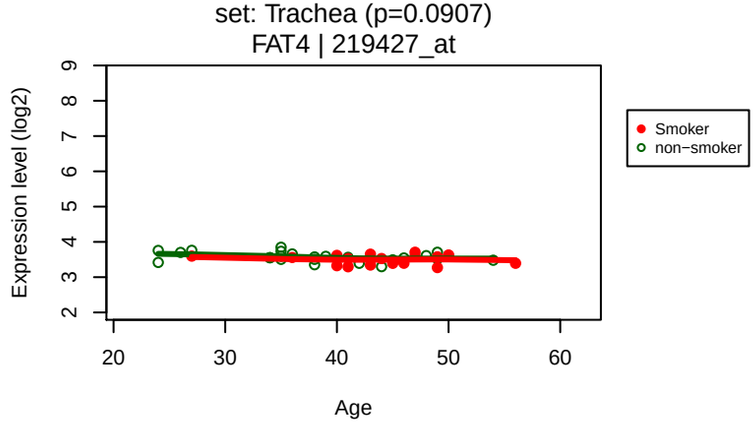

Gene expression ratio in Males
gene: FAT4 | 219427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9338
p=0.1114
p=0.4321
p=0.9109
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAT4 | 219427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4809
p=0.3533
p=0.1887
p=0.7272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

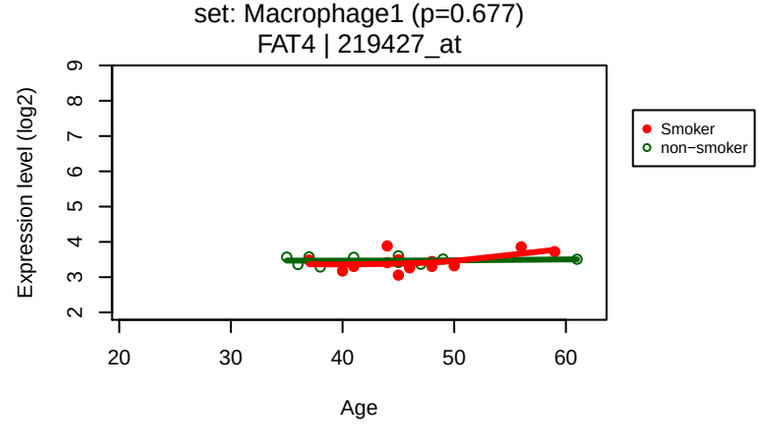

set: SmallAirway1 (p=0.089)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker

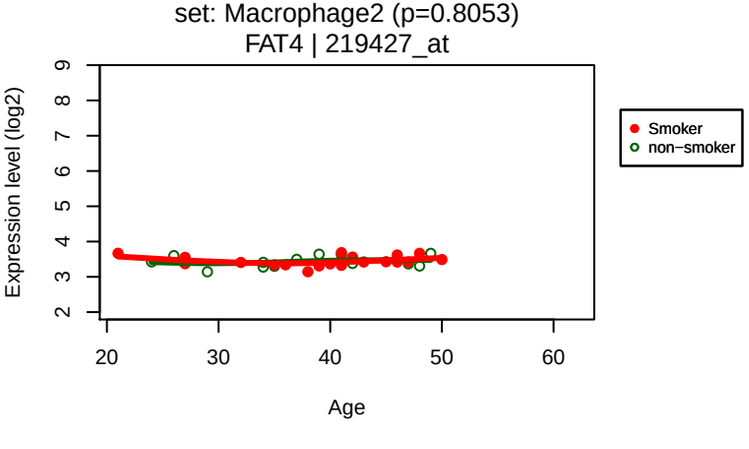

set: SmallAirway2 (p=0.55)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker

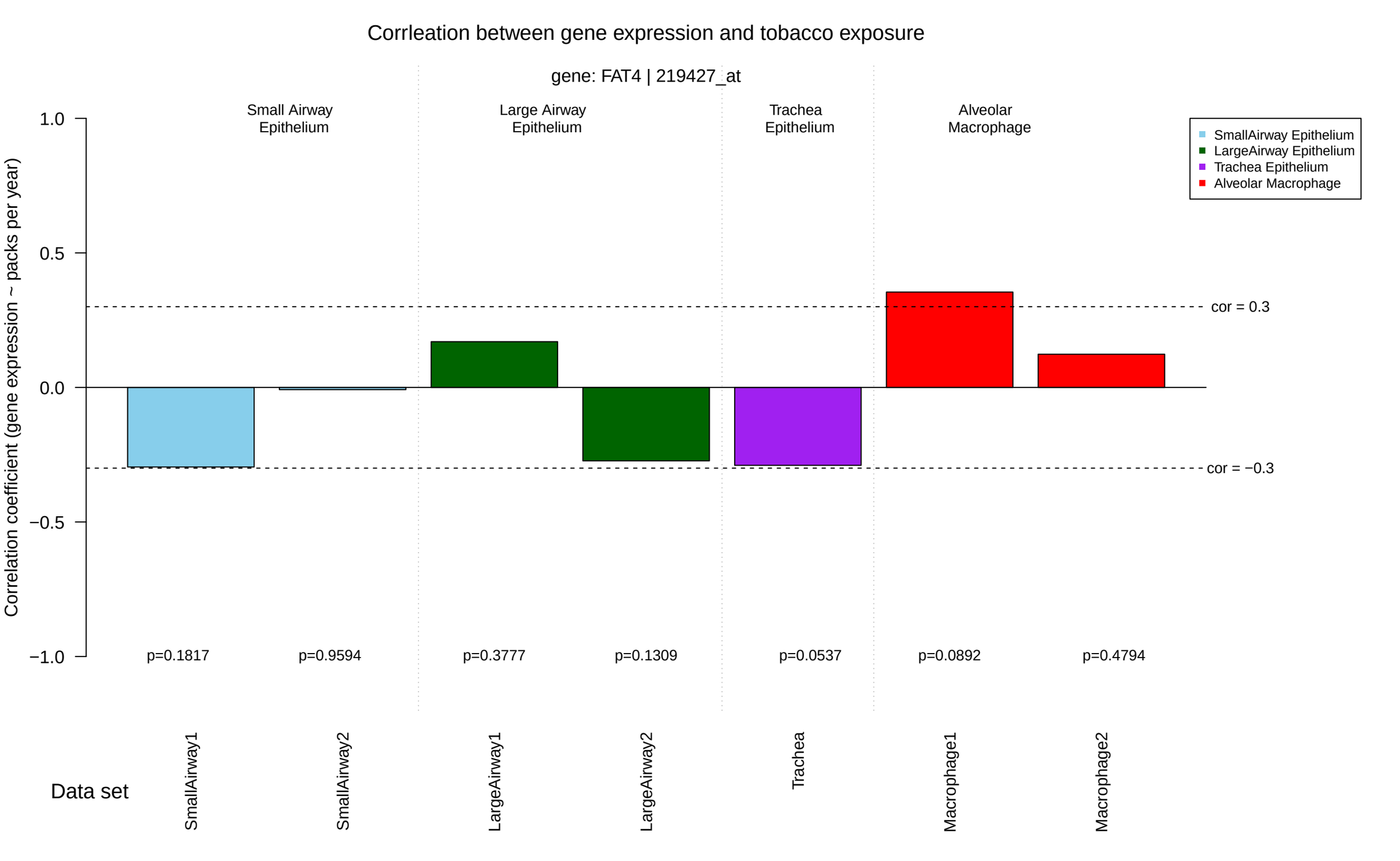

set: LargeAirway1 (p=0.2817)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3976)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0907)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.677)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8053)
FAT4 | 219427_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAT4 | 219427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1817
p=0.9594
p=0.3777
p=0.1309
p=0.0537
p=0.0892
p=0.4794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2